Homework 01 Solutions

Author: Dustin Miao (dustinmi)
Course: 10-783 A
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Task 1.1

We have that X is a zero-centered Gaussian since e^{-t} is a deterministic function of time. Then, by Ito isometry,

$$\mathrm{Var}(X) = \mathbf{E}\left[\left(\int_0^2 e^{-t}\,\mathrm{d}W_t\right)^2\right] = \int_0^2 e^{-2t}\,\mathrm{d}t = \frac{1-e^{-4}}{4}.$$

Thus, $X \sim \mathcal{N}(0, (1-e^{-4})/4)$.

Task 1.2

Let $f(t)$ be the piecewise function

$$f(t) = \begin{cases} 1 & 0 \le t \le 1/2 \\ 2 & 1/2 \le t \le 1 \end{cases}.$$

Because $f(t)$ is time-dependent, X is a zero-centered Gaussian with variance

$$\mathrm{Var}(X) = \mathbf{E}\left[\left(\int_0^1 f(t)\,\mathrm{d}W_t\right)^2\right] = \int_0^1 f(t)^2\,\mathrm{d}t = 2.5.$$

Thus, $X \sim \mathcal{N}(0, 2.5)$.

Task 1.3

We have that

$$\begin{aligned}
\mathrm{Var}(X) &= \mathbf{E}\left[\left(\int_0^1 t\,\mathrm{d}W_t\right)^2\right] = \int_0^1 t^2\,\mathrm{d}t = 1/3 \\
\mathrm{Var}(Y) &= \mathbf{E}\left[\left(\int_0^1 (1-t)\,\mathrm{d}W_t\right)^2\right] = \int_0^1 (1-t)^2\,\mathrm{d}t = 1/3 \\
\mathrm{Cov}(X,Y) &= \int_0^1 t(1-t)\,\mathrm{d}t = 1/6.
\end{aligned}$$

Thus, X and Y are not independent because they have a non-zero covariance.

Task 1.4

Let $a, b \in \mathbb{R}$ be arbitrary reals. Then,

$$aX + bY = a \int_0^1 1 \, \mathrm{d}W_t + b \int_0^1 \left(t - \frac{1}{2}\right) \, \mathrm{d}W_t = \int_0^1 \left(a + bt - \frac{b}{2}\right) \, \mathrm{d}W_t.$$

Since the integrand is deterministic, it follows that the distribution of $aX + bY$ must be Gaussian, showing that the joint distribution is Gaussian. Then,

$$\mathrm{Cov}(X, Y) = \int_0^1 \left(t - \frac{1}{2}\right)^2 \, \mathrm{d}t = \frac{1}{3}\left(t - \frac{1}{2}\right)\Big|_0^1 = 0.$$

Since the joint distribution is Gaussian and the covariance is 0, it follows that X and Y are independent.

Task 1.5

The integral form of Ito's formula is

$$f(W_t) - f(W_0) = \int_0^T f'(W_t) \, \mathrm{d}W_t + \frac{1}{2} \int_0^T f''(W_t) \, \mathrm{d}t.$$

Substituting $f(x) = x^3/3$ and rearranging gives

$$\int_0^1 W_t^2 \, \mathrm{d}W_t = \frac{1}{3} W_1^3 - \int_0^1 W_t \, \mathrm{d}t.$$

Task 1.6

(a) Evaluating Ito's formula with $f(x) = \frac{1}{2}x^2$ in the interval $[0, 1]$ gives

$$\frac{1}{2} W_1^2 = \int_0^1 W_t \, \mathrm{d}W_t + \frac{1}{2}.$$

Rearranging and taking the expectation, we get that

$$\mathbf{E}\left[\int_0^1 W_t \, \mathrm{d}W_t\right] = \mathbf{E}\left[\frac{W_1^2 - 1}{2}\right] = \frac{\mathbf{E}\left[W_1^2\right] - 1}{2} = 0.$$

(b) From Ito Isometry, we have that

$$\mathbf{E}\left[\left(\int_0^1 W_t \, \mathrm{d}W_t\right)^2\right] = \mathbf{E}\left[\int_0^1 W_t^2 \, \mathrm{d}t\right] = \int_0^1 \mathbf{E}\left[W_t^2\right] \, \mathrm{d}t = \int_0^1 t \, \mathrm{d}t = 1/2.$$

Task 1.7

Consider the product rule applied to processes W_t and t on $[0, 1]$:

$$W_1 = \int_0^1 W_t \, \mathrm{d}t + \int_0^1 t \, \mathrm{d}W_t + \int_0^1 \mathrm{d}[W_t, t].$$

The final term on the RHS reduces to 0, so rearranging gives the desired result of

$$\int_0^1 t \, \mathrm{d}W_t = W_1 - \int_0^1 W_t \, \mathrm{d}t.$$

Then,

$$\begin{aligned}
\mathrm{Var}\left(\int_0^1 W_t \, \mathrm{d}t\right) = \mathbf{E}\left[\left(\int_0^1 W_t \, \mathrm{d}t\right)^2\right] &= \mathbf{E}\left[\left(W_1 - \int_0^1 t \, \mathrm{d}W_t\right)^2\right] \\
&= \mathbf{E}\left[\left(\int_0^1 1 \, \mathrm{d}W_t - \int_0^1 t \, \mathrm{d}W_t\right)^2\right] \\
&= \mathbf{E}\left[\left(\int_0^1 (1-t) \, \mathrm{d}W_t\right)^2\right] \\
&= \int_0^1 (1-t)^2 \, \mathrm{d}t \\
&= 1/3.
\end{aligned}$$

Task 1.8

Let $Z = \int_a^b t \, \mathrm{d}W_t$. Z is a 0-centered Gaussian with variance

$$\mathrm{Var}(Z) = \mathbf{E}\left[\left(\int_a^b t \, \mathrm{d}W_t\right)^2\right] = \int_a^b t^2 \, \mathrm{d}W_t = \frac{1}{3}(b^3 - a^3).$$

Thus, $X \sim \mathcal{N}(0, 1/24)$, $Y \sim \mathcal{N}(0, 7/24)$, and

$$X + Y = \int_0^1 t \, \mathrm{d}W_t \sim \mathcal{N}(0, 1/3).$$

Then, X and Y are independent since they are integrals of deterministic function against Brownian increments $[0, 1/2]$ and $[1/2, 1]$, which are independent.

Task 1.9

Since the integrand of X_s is deterministic, it must have mean 0: $\mathbf{E}[X_S] = 0$. Then, the covariance of X_S and X_t is given by

$$\mathrm{Cov}(X_s, X_t) = \mathbf{E}\left[\left(\int_0^s u \, \mathrm{d}W_u\right)\left(\int_0^t u \, \mathrm{d}W_u\right)\right] = \int_0^{\min(s,t)} u^2 \, \mathrm{d}u = \frac{1}{3}(\min(s, t))^3.$$

It follows that the variance $\mathrm{Var}(X_t) = \frac{1}{3}t^3$.

Let $s_1, \ldots, s_n > 0$ and $a_1, \ldots, a_n \in \mathbb{R}$ be arbitrary time points and real coefficients, respectively. Then,

$$\sum_{i=1}^{n} a_i X_{s_i} = \sum_{i=1}^{n} \left(a_i \int_0^{s_i} t \, \mathrm{d}W_t \right) = \int_0^1 \left(\sum_{i=1}^{n} a_i t \mathbf{1}_{t \le s_i} \right) \, \mathrm{d}W_t.$$

The integrand is deterministic, so the integral must be Gaussian. Thus, the join distribution is Gaussian, showing that $(X_s)_{s\in[0,1]}$ is a Gaussian process.

Task 1.10

(a) True. Since $f \in L^2([0,1])$, by the Density of Step Function theorem, we have that for some sequence of step functions $f_n \in L^2(()[0,1])$

$$f_n(t) = \sum_{i=0}^{k_n - 1} c_i^{(n)} \mathbf{1}_{[t_i, t_{i+1}]}(t),$$

that $f_n \xrightarrow{L_2} f$. Define $X_n = \int_0^1 f_n(t) \, \mathrm{d}W_t$ and $X = \int_0^1 f(t) \, \mathrm{d}W_t$. From Ito isometry,

$$\mathbf{E}\left[(X_n - X)^2\right] = \mathbf{E}\left[\left(\int_0^1 (f_n(t) - f(t)) \, \mathrm{d}W_t \right)^2 \right] = \int_0^1 (f_n(t) - f(t))^2 \, \mathrm{d}t,$$

which tends to 0 as $n \to \infty$. This shows that $X_n \xrightarrow{L_2} X$. Furthermore, for any X_n, we have that

$$X_n = \int_0^1 f_n(t) \, \mathrm{d}W_t = \sum_{i=0}^{k_n - 1} c_i^{(n)} (W_{t_{i+1}} - W_{t_i}).$$

Since $[t_i, t_{i+1}]$ interval is disjoint and Wiener process increments are normally distributed, this means that each X_n is a weighted sum of independent Gaussians and is thus a Gaussian random variable. Since each X_n is Gaussian, $X_n \xrightarrow{L_2} X$, and Gaussian distributions are closed under convergence in distribution, it follows that X is also a Gaussian random variable, as desired.

(b) False. For $X_t = W_t$ (adapted and in $L^2([0,1])$, the integral evaluates to $\frac{1}{2}(W_1^2 - 1)$ which is a shifted and scaled χ^2 variable.

(c) True. For an adapted, square-integratable stochastic process X_t, the Ito integral $M_t = \int_0^t X_s \, \mathrm{d}W_t$ is a true martingale w.r.t. the filtration of Brownian motion. Then, by the martingale property, the expected value remains constant over time, so $\mathbf{E}[M_1] = \mathbf{E}[M_0] = 0$.

(d) True. For any deterministic $f \in L^2([0,1])$, the Ito integral has mean 0 (from part c, since any deterministic function must be adapted). Then, by Ito isometry,

$$\begin{aligned}
\mathrm{Var}\left(\int_0^1 f(t) \, \mathrm{d}W_t \right) &= \mathbf{E}\left[\left(\int_0^1 f(t) \, \mathrm{d}W_t \right) \right] - \mathbf{E}\left[\int_0^1 f(t) \, \mathrm{d}W_t \right]^2 \\
&= \mathbf{E}\left[\left(\int_0^1 f(t) \, \mathrm{d}W_t \right)^2 \right] \\
&= \int_0^1 f(t)^2 \, \mathrm{d}t,
\end{aligned}$$

as desired.

Task 2.1

We have for any Π_n that

$$\begin{aligned}
Q_{\Pi_n}(X_t) &= \sum_{i=0}^{n-1}(X_{t_{i+1}} - X_{t_i})^2 \\
&= \sum_{i=0}^{n-1}(W_{t_{i+1}} - W_{t_i} + f(t_{i+1}) - f(t_i))^2 \\
&= \sum_{i=0}^{n-1}(W_{t_{i+1}} - W_{t_i})^2 + \sum_{i=0}^{n-1}(f(t_{i+1}) - f(t_i))^2 + 2\sum_{i=0}^{n-1}(W_{t_{i+1}} - W_{t_i})(f(t_{i+1}) - f(t_i)) \\
|Q_{\Pi_n}(X_t) - Q_{\Pi_n}(W_t)| &\leq Q_{\Pi_n}(f) + 2\left|\sum_{i=0}^{n-1}(W_{t_{i+1}} - W_{t_i})(f(t_{i+1}) - f(t_i))\right| \\
&\leq Q_{\Pi_n}(f) + 2\sum_{i=0}^{n-1}\left|W_{t_{i+1}} - W_{t_i}\right|\left|f(t_{i+1}) - f(t_i)\right| \\
&\leq Q_{\Pi_n}(f) + 2\sqrt{\left(\sum_{i=0}^{n-1}(W_{t_{i+1}} - W_{t_i})^2\right)\left(\sum_{i=0}^{n-1}(f(t_{i+1}) - f(t_i))^2\right)} \\
&= Q_{\Pi_n}(f) + 2\sqrt{Q_{\Pi_n}(W)Q_{\Pi_n}(f)}
\end{aligned}$$

We have that $Q_{\Pi_n}(W_t) \xrightarrow{L_2} T$ and $Q_{\Pi_n}(f) \xrightarrow{L_2} 0$. The bound shows that $|Q_{\Pi_n}(X_t) - Q_{\Pi_n}(W_t)| \xrightarrow{L_2} 0$ (since $Q_{\Pi_n}(f)$ is deterministic), so $Q_{\Pi_n}(X_t) \xrightarrow{L_2} T$ as well.

Task 2.2

We show by contradiction that a strictly positive quadratic variance implies that the total variance must be infinite. Assume the contrary. Then,

$$\begin{aligned}
Q_{\Pi_n}(X_t) &= \sum_{i=0}^{n-1}(X_{t_{i+1}} - X_{t_i})^2 \\
&= \sum_{i=0}^{n-1}\left|X_{t_{i+1}} - X_{t_i}\right|^2 \\
&\leq \omega_X(\Delta t)\sum_{i=0}^{n-1}\left|X_{t_{i+1}} - X_{t_i}\right| \\
&= \omega_X(\Delta t)\mathrm{TV}(X)
\end{aligned}$$

Assuming X is almost-surely continuous, the total variation is finite and the modulus of continuity tends to 0 as $\Delta t \to 0$, therefore the quadratic variation must be 0, a contradiction.

Since we showed in Task 2.1 that the quadratic variation is T for any time $T > 0$, it follows that the total variation must be infinite, as desired.

Task 3

(1)

t	E[W_n(t)]	theory=0	Var[W_n(t)]	theory=t
0.100	-0.00473	0.000	0.10167	0.100
0.250	-0.00961	0.000	0.25102	0.250
0.500	-0.01644	0.000	0.50755	0.500
0.750	-0.02026	0.000	0.77095	0.750
1.000	-0.02012	0.000	1.02874	1.000

(2) See Figure 1.

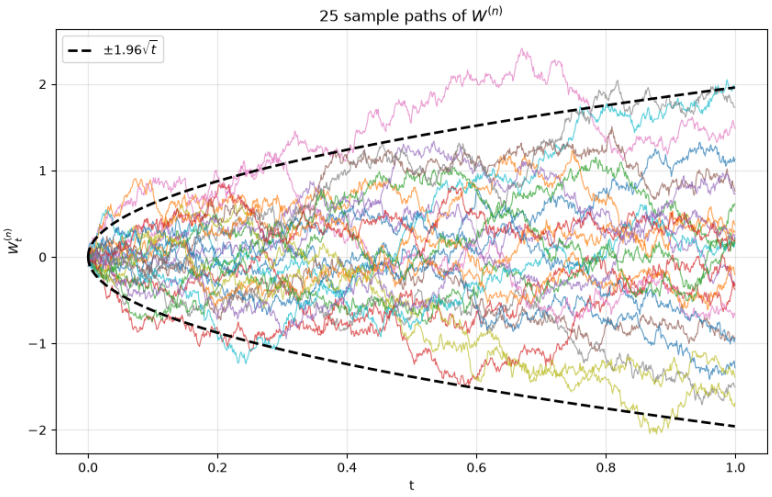


Figure 1: 25 generated Wiener processes with 95% envelope

(3) See Figure 2.

	empirical	theoretical
mean	-0.00479	0.00000
variance	0.39814	0.40000

Task 4.1

For I_1, the integrand is a deterministic function of time, so it must be Gaussian distributed with mean 0. The variance is given by Ito isometry

$$\mathrm{Var}(I_1) = \mathbf{E}\left[\left(\int_0^1 t \, \mathrm{d}W_t\right)^2\right] = \int_0^1 t^2 \, \mathrm{d}t = 1/3,$$

giving that $I_1 \sim \mathcal{N}(0, 1/3)$, as desired.

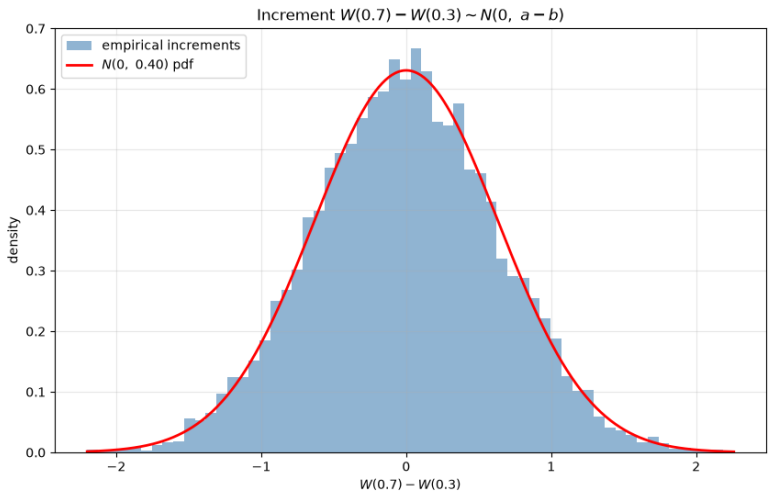


Figure 2: Empirical distribution of Wiener process increments

Task 4.2

See Figure 3. We use $m = 10{,}000$ Wiener trajectories and $n = 5000$ subintervals of $[0, 1]$, evaluating the integrand at the left endpoint. The empirical distribution is close to the theoretical $\mathcal{N}(0, 1/3)$ distribution.

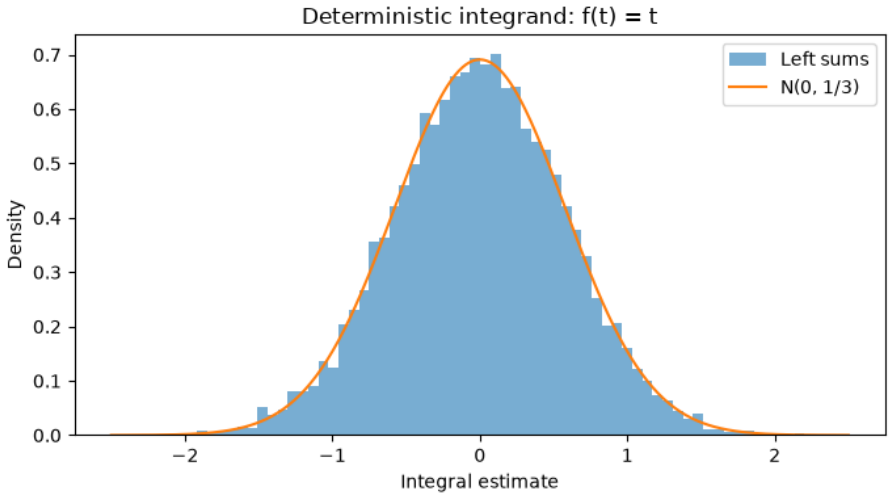


Figure 3: Empirical distribution of left tagged sums for $\int_0^1 t \, \mathrm{d}W_t$ with the theoretical Gaussian density

Task 4.3

See Figure 4. For each $n \in \{10, 20, 50, 100, 200, 500, 1000, 2000, 5000\}$, we calculate all three tagged sums on the same $m = 10{,}000$ trajectories and take the maximum absolute differences across trajectories. We have that

$$S_n(t, \texttt{mid}) - S_n(t, \texttt{left}) = \frac{W_1}{2n},$$
$$S_n(t, \texttt{right}) - S_n(t, \texttt{left}) = \frac{W_1}{n}.$$

Thus, both differences converge to 0, so the tag choice does not affect the limiting integral.

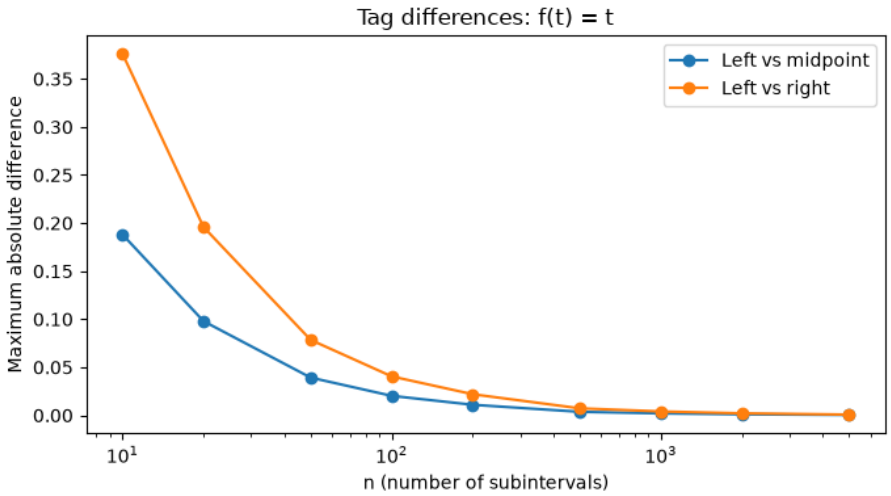


Figure 4: Maximum absolute left–midpoint and left–right tag differences for the deterministic integrand $f(t) = t$

Task 4.4

See Figure 5. We again use $m = 10{,}000$ trajectories and $n = 5000$ subintervals, now with $f(t) = W_t$ and left endpoint tags. By Ito's formula,

$$I_2 = \int_0^1 W_t \, \mathrm{d}W_t = \frac{W_1^2 - 1}{2}.$$

Thus, the limiting distribution is a shifted and scaled χ_1^2 distribution, rather than a Gaussian, consistent with the histogram. We use $\mathrm{d}W_t$ as specified in the definition of the stochastic sums and the caption of Figure 3 in the handout.

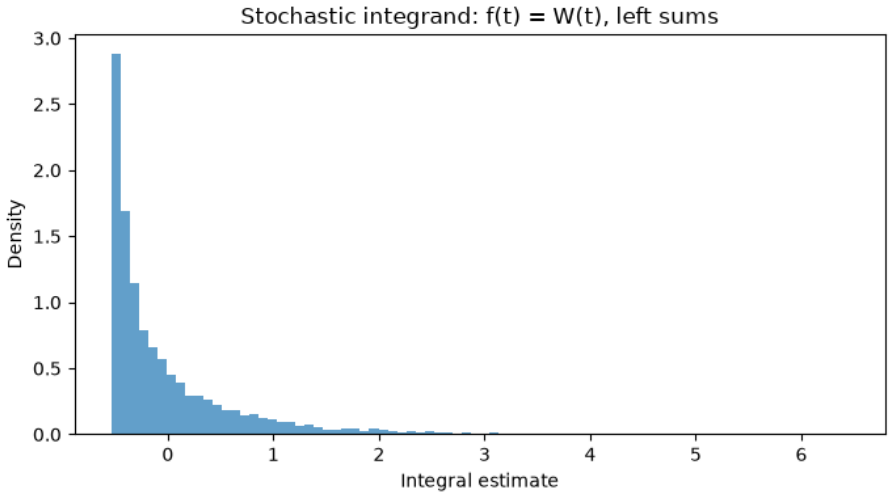


Figure 5: Empirical distribution of left tagged sums for $\int_0^1 W_t \, \mathrm{d}W_t$

Task 4.5

See Figure 6. We repeat Task 4.3 with $f(t) = W_t$, using the same values of n and m. The midpoint tags use the Brownian values at the midpoint times, obtained by simulating on a grid with $2n$ steps. The limits are

$$S_n(W, \texttt{left}) \to \frac{W_1^2 - 1}{2},$$
$$S_n(W, \texttt{mid}) \to \frac{W_1^2}{2},$$
$$S_n(W, \texttt{right}) \to \frac{W_1^2 + 1}{2}.$$

The left–midpoint and left–right differences approach $1/2$ and 1, respectively. Thus, the tag choice matters for this stochastic integrand.

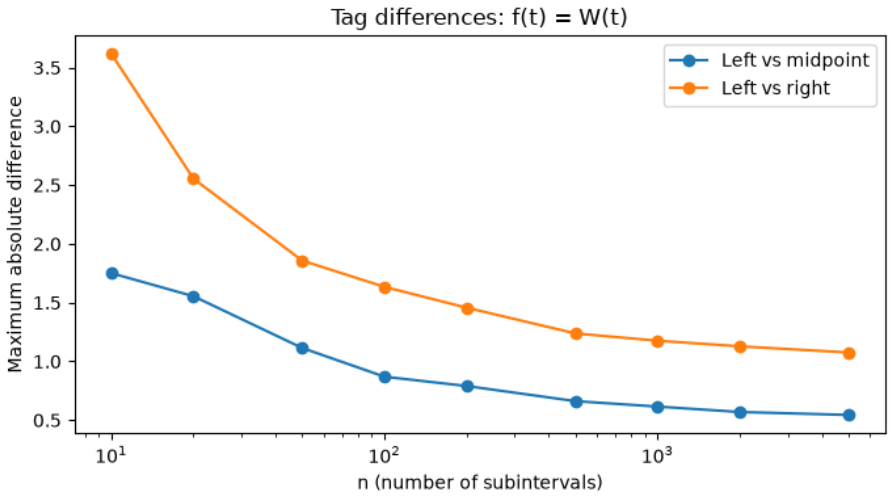


Figure 6: Maximum absolute left–midpoint and left–right tag differences for the stochastic integrand $f(t) = W_t$

Collaboration Questions

(1) (a) Did you receive any help whatsoever from anyone in solving this assignment? No.
    (b) If you answered 'yes', give full details. N/A

(2) (a) Did you give any help whatsoever to anyone in solving this assignment? No.
    (b) If you answered 'yes', give full details. N/A.

(3) (a) Did you find or come across code that implements any part of this assignment? No.
    (b) If you answered 'yes', give full details. N/A